

# Exact finite-frame inference under one-start systematic probability-proportional-to-size sampling

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## Abstract

The ordinary Stringer upper bound is routinely paired with monetary-unit sampling, although a one-start systematic probability-proportional-to-size (PPS) sample is dependent rather than i.i.d. We determine what can be guaranteed under the systematic design itself. For every sample size  $n \geq 2$  and every tail level  $\alpha \in (0, 1)$ , a periodic two-item-per-cell population makes binomial-factor Stringer undercover; the analogous Poisson rule also fails whenever  $-\log(\alpha)/n < 1 - \alpha$ . The systematic sample mean nevertheless remains exactly design-unbiased. This identity implies both a sharp impossibility result and a remedy: every universally valid taint-only bound must be at least  $1 - \alpha + \alpha y$  on the constant sample  $(y, \dots, y)$ , and that floor is attained. Intersecting it with a deterministic item-completion bound and then taking the maximum with capped Stringer yields a design-valid overlay. When the ordered frame is known, exact inversion of the grid-mean randomization rank gives a pointwise smaller least-favourable-completion bound, expressible as finitely many rational linear programs. For phase-partitioned frames this optimization reduces to a minimum-coalition formula; equal phases give a closed form, rational unequal phases admit an  $O(KD)$  pseudo-polynomial algorithm, and exact evaluation is weakly NP-hard. Independent starts yield an explicit all-zero formula. In a 2,000-item, 100-draw example, exact inversion lowers the valid fallback from 0.95 to 0.90.

**Keywords:** monetary-unit sampling; systematic PPS; Stringer bound; finite-population inference; randomization inference; linear programming; subset sum.

## 1 Introduction

Monetary-unit sampling selects book units with probability proportional to recorded amount. A common implementation places the ordered population on the cumulative-book-value line, chooses one random start, and takes equally spaced points thereafter. This one-start systematic PPS design is easy to implement, but the observations within its grid are dependent. Professional guidance permits systematic selection with random starts [6], and statistical treatments of monetary-unit sampling emphasize the importance of the actual unequal-probability design [2].

For comparison, let  $t_{(1)} \geq \dots \geq t_{(n)}$  be observed taints in nonincreasing order. At tail

level  $\alpha$ , the ordinary Stringer calculation is

$$\text{SB}_\alpha = p_n(0) + \sum_{j=1}^n \{p_n(j) - p_n(j-1)\}t_{(j)}. \quad (1)$$

For the binomial convention,  $p_n(j)$  is the upper Clopper–Pearson limit: for  $j < n$  it is the solution of

$$\sum_{i=0}^j \binom{n}{i} p^i (1-p)^{n-i} = \alpha, \quad p_n(n) = 1.$$

For the Poisson convention, if  $\lambda_j$  solves

$$e^{-\lambda_j} \sum_{i=0}^j \frac{\lambda_j^i}{i!} = \alpha,$$

then  $p_n(j) = \min(1, \lambda_j/n)$ . These factor sequences arise from independent count models; applying them to one systematic grid therefore requires a separate design argument. Audit tables have long used the Poisson convention [1], while Stringer’s original construction uses binomial limits [7].

Prior work already warns against identifying systematic selection with simple random sampling. Hoogduin, Hall, Tsay, and Pierce [5] showed by simulation that this mismatch can materially distort assessed risk. Fienberg, Neter, and Leitch [3] obtained finite-sample audit bounds by maximizing over a multinomial confidence region. Their discretized multinomial model differs from the fixed ordered-frame randomization problem studied here, but it is an important optimization-based predecessor. Recent work on randomized systematic unequal-probability sampling derives exact second-order inclusion probabilities for variance estimation [8]; its target is different from the one-sided taint-completion problem here. Thus the contribution below is not a first warning about systematic sampling. It is an exact analysis: an all- $n$ , all-confidence binomial failure family, the sharp information limit for taint-only bounds, a closed-form valid overlay, and a finite-frame randomization inversion.

The main mechanism is the distinction between first moments and joint dependence. Averaging over the random start makes the systematic grid mean exactly unbiased for the finite-population taint ratio. A periodic ordering can nevertheless make every point in the grid reveal the same phase. The first fact yields a Markov bound; the second proves that its value on constant samples cannot be improved without using item identities or frame structure. Once the complete ordered frame is available, the unobserved taints become nuisance variables in an exact randomization test. Maximizing the population ratio over compatible completions produces the sharper finite-frame bound.

Sections 2 and 3 develop these two levels. Section 4 identifies the phase-partitioned case in which the general linear programs collapse to subset sum and derives the independent-start formula. Section 5 states the role of the exact finite computations. The universal theorems assume a fixed ordered population with positive known book weights, taints in  $[0, 1]$ , and a uniform random start; they do not cover successive PPS, stratification, nonrandom starts, uncertain book amounts, or negative taints.

## 2 One-start design: unbiasedness, failure, and a sharp safeguard

The key split is between first-moment calibration and dependence: the sample average is design-unbiased, but the complete grid can behave as a single random observation. We first prove that separation, then derive the resulting failure family and use the same first-moment identity to construct a closed-form valid reporting rule.

### 2.1 Finite-population model and the two-phase construction

Let an ordered finite population have positive recorded amounts  $w_1, \dots, w_N$  and taints  $t_1, \dots, t_N \in [0, 1]$ . Put

$$W = \sum_{i=1}^N w_i, \quad \theta = \frac{1}{W} \sum_{i=1}^N w_i t_i.$$

Place the items consecutively on  $[0, W)$  and let  $\tau(u)$  equal the taint of the item containing  $u$ . For intended sample size  $n$ , write  $h = W/n$ , draw  $R \sim \text{Unif}[0, h)$ , and observe

$$T_k = \tau(R + kh), \quad k = 0, \dots, n-1.$$

Endpoints have probability zero and may be assigned by any fixed convention. This is the continuous book-unit form of one-start systematic PPS; the integer-book-unit version chooses one of the positions in an integer interval.

**Proposition 1** (Design-unbiased systematic mean). *For every such ordered population,*

$$\mathbb{E}_R \bar{T} = \theta, \quad \bar{T} = \frac{1}{n} \sum_{k=0}^{n-1} T_k.$$

*Proof.* Integration over the systematic cells gives

$$\mathbb{E}_R \bar{T} = \frac{1}{n} \sum_{k=0}^{n-1} \frac{1}{h} \int_0^h \tau(r + kh) dr = \frac{1}{W} \int_0^W \tau(u) du = \theta.$$

□

Unbiasedness does not imply a binomial count law. Fix  $0 \leq y < 1$  and  $0 < q < 1$ . In each of the  $n$  systematic cells put first an item of weight  $qh$  and taint one, then an item of weight  $(1-q)h$  and taint  $y$ . The complete sample is

$$(T_0, \dots, T_{n-1}) = \begin{cases} (1, \dots, 1), & \text{with probability } q, \\ (y, \dots, y), & \text{with probability } 1 - q, \end{cases} \quad \theta = y + q(1 - y). \quad (2)$$

All items are smaller than one systematic interval, so this construction does not depend on duplicate-hit or certainty-item conventions.

**Theorem 2** (Exact systematic-PPS failure family). *For every  $n \geq 2$  and every  $\alpha \in (0, 1)$ , some ordered finite population of the form (2) gives ordinary binomial Stringer coverage strictly below  $1 - \alpha$ . For ordinary Poisson factors the same conclusion holds whenever*

$$\frac{-\log \alpha}{n} < 1 - \alpha. \quad (3)$$

*Thus Poisson Stringer fails on some population at 90%, 95%, and 99% confidence for every  $n \geq 3, 4, 5$ , respectively. Final capping at one does not alter these conclusions.*

*Proof.* For binomial factors,  $p_n(0) = 1 - \alpha^{1/n}$  and telescoping the increments on the low phase gives

$$\text{SB}_B(y, \dots, y) = p_n(0) + \{1 - p_n(0)\}y = 1 - \alpha^{1/n}(1 - y).$$

Because  $\alpha^{1/n} > \alpha$ , choose

$$1 - \alpha^{1/n} < q < 1 - \alpha.$$

Then the low-phase bound is below  $\theta$ , while that phase has probability  $1 - q > \alpha$ . This proves binomial undercoverage. The high-phase binomial bound is one, so coverage in this family is exactly  $q$ .

For Poisson factors take  $y = 0$ . The all-zero bound is  $\lambda_0/n = (-\log \alpha)/n$ . Under (3), choose  $(-\log \alpha)/n < q < 1 - \alpha$ . Again the low phase is a noncoverage event of probability greater than  $\alpha$ . The stated integer thresholds follow by direct substitution. In both arguments the open interval for  $q$  contains a rational  $a/m$ ; taking systematic interval  $h = m$  gives the same construction with integer book amounts  $a$  and  $m - a$  in every cell.  $\square$

The smallest exact witness in a unit-item census at 95% confidence has only nine book units. With  $n = 3$ , interval three, and ordered taints

$$(1, 1, 0 \mid 1, 1, 0 \mid 1, 1, 0),$$

the three starts return three ones, three ones, and three zeros. The target is  $2/3$ , whereas  $p_3(0) = 1 - 0.05^{1/3} < 2/3$ , so exact coverage is  $2/3$  rather than  $0.95$ . Exact enumeration of all 2,188 ordered binary populations and 4,444 phases through total size nine finds no smaller unit-item witness; this finite minimality statement is limited to that enumerated class.

A practical-size version makes the failure scale explicit. Take total recorded amount 1,000,  $n = 100$ , and interval 10. Repeat 100 times an item of amount one and taint one followed by an item of amount nine and taint zero. One start gives 100 ones and the other nine starts give 100 zeros, while  $\theta = 0.1$ . On the high phase the binomial bound is one. For Poisson factors, the exact inequality  $\mathbb{P}\{\text{Pois}(10) \leq 10\} > 0.10$  implies  $\lambda_{100} > 10$ , so that phase also covers at all three levels below. Consequently, both ordinary factor conventions have exact coverage  $0.1$ .

## 2.2 A sharp taint-only floor and an item-aware improvement

The construction also identifies the unavoidable price of robustness if a rule sees only the taint vector. Item identities and book weights will then provide a separate, strictly useful information channel.

| confidence | B zero    | P zero    | ordinary coverage | safe output | B uplift  | P uplift  |
|------------|-----------|-----------|-------------------|-------------|-----------|-----------|
| 90%        | 0.0227628 | 0.0230259 | 0.1000000         | 0.1000000   | 0.0772372 | 0.0769741 |
| 95%        | 0.0295130 | 0.0299573 | 0.1000000         | 0.1000000   | 0.0704870 | 0.0700427 |
| 99%        | 0.0450074 | 0.0460517 | 0.1000000         | 0.1000000   | 0.0549926 | 0.0539483 |

Table 1: The 100-draw periodic population. “B zero” and “P zero” are the binomial- and Poisson-factor all-zero outputs. The design-valid safe output is defined below. Displays are rounded from exact rational brackets in the systematic-PPS calculation.

**Proposition 3** (Necessary taint-only diagonal). *If  $V_\alpha : [0, 1]^n \rightarrow [0, 1]$  has systematic-design coverage at least  $1 - \alpha$  for every ordered finite population and depends only on the observed taints, then, for every  $0 \leq y < 1$ ,*

$$V_\alpha(y, \dots, y) \geq 1 - \alpha + \alpha y.$$

*Proof.* If the inequality failed, choose

$$\frac{V_\alpha(y, \dots, y) - y}{1 - y} < q < 1 - \alpha$$

and use (2). Its target  $y + q(1 - y)$  would exceed the reported low-phase value, and that phase would have probability  $1 - q > \alpha$ .  $\square$

The lower bound is attained on every diagonal by

$$M_\alpha = 1 - \alpha + \alpha \bar{T}. \quad (4)$$

Indeed, with  $X = 1 - \bar{T}$ , Proposition 1 gives  $\mathbb{E}_R X = 1 - \theta$ . The event  $M_\alpha < \theta$  is  $X > (\mathbb{E}_R X)/\alpha$ , whose probability is at most  $\alpha$  by Markov’s inequality. Thus one random start can carry only one observation’s worth of distribution-free information, regardless of the number of points in its grid.

For a sampled set  $S$  of distinct item identities, define the deterministic completion bound

$$C(S) = \frac{1}{W} \left( \sum_{i \in S} w_i t_i + \sum_{i \notin S} w_i \right).$$

Because each unobserved taint is at most one,  $C(S) \geq \theta$  pointwise. This extra information permits a smaller valid floor.

**Proposition 4** (Systematic-safe Stringer overlay). *Put*

$$H_\alpha = \min\{M_\alpha, C(S)\}, \quad \text{SB}_\alpha^{\text{sys}} = \max\{\min(1, \text{SB}_\alpha), H_\alpha\}. \quad (\text{SYS})$$

*For either the binomial- or Poisson-factor ordinary calculation,  $\text{SB}_\alpha^{\text{sys}}$  has coverage at least  $1 - \alpha$  under the one-start systematic design for every ordered finite population. It is never below the ordinary output after its natural cap at one.*

*Proof.* Whenever  $M_\alpha \geq \theta$ , the deterministic inequality  $C(S) \geq \theta$  implies  $H_\alpha \geq \theta$ . Hence  $H_\alpha$  has noncoverage at most  $\alpha$ . The maximum in (SYS) is at least  $H_\alpha$ , which proves coverage, and at least the capped ordinary output by definition.  $\square$

In the all-zero phase of Table 1, the 100 audited taint-zero items comprise 900 of the 1,000 recorded units. At 95% confidence,  $M_{0.05} = 0.95$  but  $C(S) = 0.10$ , so  $H_{0.05} = 0.10$ . The safe output is therefore exactly 0.10, an uplift of about 0.070487 over binomial Stringer and 0.070043 over Poisson Stringer, rather than the 0.92 uplift of the taint-only floor. On the all-one phase,  $H_\alpha = 1$ .

### 3 Exact finite-frame minimax inversion

The closed-form bound uses the observed identities only through  $C(S)$ . A fixed known frame contains more information: it specifies which items every other random start would have selected. We now invert that complete finite randomization distribution.

Combine random-start intervals having the same item-hit multiplicities into phase atoms  $a = 1, \dots, K$ , with probabilities  $\pi_a > 0$ . Let  $m_{ai}$  be the number of hits on item  $i$  in atom  $a$ , and for a complete candidate taint vector  $u \in [0, 1]^N$  put

$$Y_a(u) = \frac{1}{n} \sum_{i=1}^N m_{ai} u_i, \quad \theta(u) = \frac{1}{W} \sum_{i=1}^N w_i u_i.$$

Suppose the observed phase audits the distinct set  $S$ , returns values  $x = (x_i)_{i \in S}$ , and has grid mean  $y$ . Define the exact lower-tail rank

$$F_u(y) = \sum_{a=1}^K \pi_a \mathbf{1}\{Y_a(u) \leq y\}$$

and the least-favourable-completion bound

$$B_\alpha^{\text{ff}}(S, x) = \max(\{\theta(u) : 0 \leq u \leq 1, u_S = x, F_u(y) > \alpha\} \cup \{0\}). \quad (\text{FF})$$

Here “minimax” refers only to maximizing the target over nuisance-taint completions for this pre-specified grid-mean ordering. It does not assert optimality among all possible test orderings or confidence procedures.

**Proposition 5** (Exact finite-frame inversion). *For every fixed ordered frame,  $B_\alpha^{\text{ff}}$  has one-start systematic design coverage at least  $1 - \alpha$ . Pointwise,*

$$B_\alpha^{\text{ff}}(S, x) \leq \min\{M_\alpha, C(S)\} = H_\alpha. \quad (5)$$

*Moreover, for rational book weights and observed taints, the value in (FF) is the maximum of finitely many rational linear programs indexed by inclusion-minimal phase coalitions of probability strictly greater than  $\alpha$ .*

*Proof.* Fix the true population  $t$ , and let  $F_t$  be the CDF of its grid mean over the random start. If  $B_\alpha^{\text{ff}}(S_R, t_{S_R}) < \theta(t)$ , then the true completion is absent from the feasible set in (FF); therefore  $F_t(Y_R(t)) \leq \alpha$ . For any discrete random variable  $Y$ ,

$$\Pr\{F_Y(Y) \leq \alpha\} \leq \alpha,$$

because the event, when nonempty, has probability equal to the CDF at its largest included support point. This proves design coverage.

Every compatible completion is bounded by  $C(S)$ . For the other inequality, take a feasible  $u$  and view  $Y$  as the random phase mean  $Y_a(u)$ . By Proposition 1,  $\mathbb{E}Y = \theta(u)$ . If  $y < 1$ , Markov's inequality applied to  $1 - Y$  gives

$$\alpha < F_u(y) = \Pr\{1 - Y \geq 1 - y\} \leq \frac{1 - \theta(u)}{1 - y},$$

so  $\theta(u) < 1 - \alpha + \alpha y = M_\alpha$ . If  $y = 1$ , the inequality follows from the unit range. The feasible set is a finite union of compact polytopes, so its maximum is attained; this proves (5).

Finally, for a phase coalition  $A$  with  $\sum_{a \in A} \pi_a > \alpha$ , impose the linear constraints

$$0 \leq u_i \leq 1, \quad u_i = x_i \ (i \in S), \quad \sum_i m_{ai} u_i \leq ny \ (a \in A),$$

and maximize the linear objective  $\theta(u)$ . The condition  $F_u(y) > \alpha$  holds exactly when the low-mean phase set contains such a coalition. Every coalition of mass above  $\alpha$  contains an inclusion-minimal one, and deleting constraints cannot lower a maximum. Taking the maximum of these rational LP values therefore gives exactly (FF).  $\square$

Thus the sharper pre-specified overlay

$$\text{SB}_\alpha^{\text{ff-sys}} = \max\{\min(1, \text{SB}_\alpha), B_\alpha^{\text{ff}}\} \quad (\text{FF-SYS})$$

is design-valid, never below capped ordinary Stringer, and never above the earlier overlay (SYS). For rational inputs, the finite maximum can be evaluated exactly by rational linear programming; equality of feasible primal and dual objectives certifies each optimum.

## 4 Phase partitions, exact algorithms, and independent starts

The general LP representation simplifies when random starts see separate parts of the frame. Call a frame *phase-partitioned* when the supports of the distinct phase-incidence atoms are pairwise disjoint and cover all items. Design unbiasedness then has the decomposition

$$\theta(u) = \sum_{a=1}^K \pi_a Y_a(u),$$

and the unobserved phase means can be varied independently.

**Proposition 6** (Unequal- and equal-phase formulas). *Suppose phase  $j$  is observed with mean  $y$  on a phase-partitioned frame. Put*

$$\rho_j(\alpha) = \min\{\pi(A) : j \in A, \pi(A) > \alpha\}, \quad \pi(A) = \sum_{a \in A} \pi_a.$$

*Then*

$$B_\alpha^{\text{ff}}(j, y) = 1 - (1 - y)\rho_j(\alpha). \quad (6)$$

*If there are  $m$  equiprobable phases, this becomes*

$$B_\alpha^{\text{ff}}(j, y) = 1 - \frac{\lfloor m\alpha \rfloor + 1}{m}(1 - y). \quad (7)$$

*Proof.* For any feasible completion let  $A = \{a : Y_a(u) \leq y\}$ . It contains the observed phase and has  $\pi(A) > \alpha$ . Therefore

$$\theta(u) \leq y\pi(A) + 1 - \pi(A) = 1 - (1 - y)\pi(A) \leq 1 - (1 - y)\rho_j(\alpha).$$

Choose a minimizing coalition  $A_\star$ . Keep the observed taints fixed, set every taint in each other phase of  $A_\star$  equal to  $y$ , and set every taint outside  $A_\star$  equal to one. The resulting phase means are  $y$  on  $A_\star$  and one elsewhere, so the completion is feasible and attains the displayed upper bound. This proves (6). With equal phase probabilities, the least cardinality having mass strictly above  $\alpha$  is  $\lfloor m\alpha \rfloor + 1$ , proving (7).  $\square$

For rational unequal probabilities, the minimum in (6) is a minimum-excess subset sum. The next result gives both an exact algorithm and the obstruction to a bit-polynomial algorithm.

**Proposition 7** (Exact dynamic program and weak NP-hardness). *Write  $\pi_a = c_a/D$  with positive integers  $c_a$  summing to  $D$ . There is an exact algorithm for (6) using  $O(KD)$  time and  $O(D)$  memory. Under binary encoding, exact evaluation is weakly NP-hard even for  $n = 1$  and an observed taint of zero.*

*Proof.* The smallest integer coalition mass strictly above  $\alpha$  is

$$h = \lfloor \alpha D \rfloor + 1.$$

Starting with the compulsory mass  $c_j$ , list the other phase masses as  $d_1, \dots, d_{K-1}$  and form the reachable sets

$$R_0 = \{0\}, \quad R_{\ell+1} = R_\ell \cup (R_\ell + d_{\ell+1}).$$

Every optional coalition yields one reachable sum and conversely. Hence the first reachable  $s$  satisfying  $c_j + s \geq h$  gives  $\rho_j(\alpha) = (c_j + s)/D$ . A Boolean array with back-pointers implements the recurrence in the stated time and memory.

For hardness, reduce from SUBSET SUM [4]. Given positive integers  $a_1, \dots, a_k$  and target  $1 \leq T \leq A := \sum_i a_i$ , take a one-draw systematic-PPS frame with weights

$$w_0 = 1, \quad w_i = 2a_i, \quad W = 1 + 2A,$$

set  $\alpha = 2T/W$ , and observe item 0 with taint zero. Each item is then a disjoint phase of probability  $w_i/W$ . A zero-taint coalition containing item 0 has weight  $1 + 2\sum_{i \in I} a_i$  and probability strictly above  $\alpha$  exactly when  $\sum_{i \in I} a_i \geq T$ . If

$$s_\star = \min\{\sum_{i \in I} a_i : \sum_{i \in I} a_i \geq T\},$$

then Proposition 6 gives

$$B_\alpha^{\text{ff}} = 1 - \frac{1 + 2s_\star}{W}.$$

Thus the original instance has a subset summing exactly to  $T$  if and only if the exact bound equals  $1 - (1 + 2T)/W$ . The reduction has polynomial encoding length. Together with the pseudo-polynomial dynamic program, this proves the claim.  $\square$

Formula (7) gives a practical-size strict improvement without solving an LP. Take 2,000 unit items,  $n = 100$ , and systematic interval 20. The 20 equiprobable phases select disjoint sets of 100 items. After one all-zero phase at 95% confidence,

$$M_{0.05} = C(S) = H_{0.05} = 0.95.$$

Here  $\lfloor 20(0.05) \rfloor + 1 = 2$ , so

$$B_{0.05}^{\text{ff}} = \frac{1800}{2000} = 0.90. \quad (8)$$

The general LP solver independently enumerates all  $\binom{20}{2} = 190$  minimal coalitions and returns the same witness with lower-tail mass 0.10; the specialized dynamic program returns the same minimum mass directly. Both ordinary all-zero Stringer outputs are below 0.03, so the finite-frame overlay reports 0.90 under either factor convention, five percentage points below the previous valid fallback.

#### 4.1 Independent starts

The disjoint equal-phase case also gives an exact replication law rather than only a generic bounded-mean reduction.

**Proposition 8** (Independent-start all-zero formula). *Use  $r \geq 1$  independent starts on  $m$  equiprobable disjoint phases, and invert the lower-tail rank of the mean of the  $r$  grid averages. Suppose all observed taints are zero and the starts visit  $d$  distinct phases. Put*

$$q_{m,r,\alpha,d} = \max\{d, \lfloor m\alpha^{1/r} \rfloor + 1\}.$$

Then

$$B_{\alpha,r}^{\text{ff}} = 1 - \frac{q_{m,r,\alpha,d}}{m}. \quad (9)$$

*Proof.* For a candidate completion, let  $Z$  be the set of zero-mean phases. Since taints are nonnegative, the mean of  $r$  independently selected phase means is at most its observed value zero exactly when every selected phase lies in  $Z$ . Its lower-tail probability is therefore  $(|Z|/m)^r$ . Compatibility forces all  $d$  observed phases into  $Z$ , and feasibility requires this probability to exceed  $\alpha$ . Thus  $|Z| \geq q_{m,r,\alpha,d}$ . Such phases contribute zero to the target and the others contribute at most  $1/m$  each, which proves the upper bound. Assigning taint zero to exactly  $q_{m,r,\alpha,d}$  phases containing the observed ones, and taint one to all remaining phases, attains equality.  $\square$

For  $m = 20$  and  $\alpha = 0.05$ , distinct all-zero starts give exact bounds 0.90, 0.75, and 0.60 for  $r = 1, 2, 3$ . For unequal rational phases, the same dynamic program replaces its one-start threshold by the least integer  $h$  satisfying  $(h/D)^r > \alpha$  and makes every distinct observed phase compulsory.

If  $r$  independent random starts are used, their  $r$  grid averages are i.i.d. variables in  $[0, 1]$  with mean  $\theta$ . Any valid distribution-free bounded-mean procedure may be applied to those averages. The effective replication is the number of independent starts, not the number of points within one grid.

## 5 Exact calculations and mathematical scope

The universal failure family, unbiasedness identity, sharp taint-only floor, valid overlays, finite-frame inversion, phase-partition formulas, complexity result, and independent-start formula are ordinary mathematical proofs. No finite experiment is extrapolated to establish those statements.

Exact rational programs establish the finite claims reported above. The nine-unit calculation enumerates all 2,188 ordered binary unit-item populations and all 4,444 associated phases in its stated search class. The 100-draw calculation uses rational phase probabilities and rigorous brackets for the binomial and Poisson factors. For the 2,000-item example, the general finite-frame solver enumerates the 190 inclusion-minimal coalitions and certifies each rational LP optimum by matching feasible primal and dual objectives; a separate denominator-cleared subset-sum program gives the same minimum coalition mass. These computations certify only the displayed census and examples. The programs use exact rational arithmetic apart from outward enclosures of transcendental confidence factors.

The sampling-design scope is also exact. One-start results average over one uniform start on a fixed ordered frame. The multiple-start proposition changes the design to independent starts and, in its closed form, assumes equiprobable disjoint phases and an all-zero observation. Alternative test orderings, overlapping-phase multiple-start formulas, stratification, successive PPS, nonrandom starts, uncertain book amounts, negative taints, and different audit estimands require separate arguments.

## 6 Discussion

The first lesson is that unbiasedness and replication are different properties. Averaging over the random start recovers the finite-population taint ratio, yet a periodic ordering can make all  $n$  observed taints encode one phase. Consequently no i.i.d. coverage theorem for the ordinary Stringer calculation could by itself validate its use under one-start systematic PPS. The reusable statistical object is the mean of each independently randomized grid.

The second lesson is an exact value-of-information statement. If the rule sees only the taint vector, every universally design-valid procedure must pay the one-observation diagonal floor  $1 - \alpha + \alpha y$ . Sampled item identities and book weights can lower that floor because they determine how much of the total recorded amount has already been resolved. The complete ordered frame contributes still more: it turns every possible start into a constraint on nuisance-taint completions. Randomization-rank inversion combines these constraints and is pointwise no larger than both closed-form fallbacks.

The phase-partition formula isolates the combinatorial core of this inversion. Disjoint phase supports make the target a weighted average of independently adjustable phase means, so inference becomes the problem of finding the least phase mass strictly above  $\alpha$ . Equal phases expose the discrete correction in closed form; unequal rational phases connect the exact bound to subset sum, explaining both the pseudo-polynomial algorithm and the weak NP-hardness boundary. With independent starts, the required zero-phase mass rises from  $\alpha$  to  $\alpha^{1/r}$ . These formulas make the inferential gain from additional random starts quantitative rather than merely qualitative.

## Data and code availability

Source code and the exact calculation data used in the finite examples are available in the project repository at [github.com/RichieSater/stringer-bound](https://github.com/RichieSater/stringer-bound).

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